

Vamsi Krishna Sai Kavety (Software Engineer, Automotive Embedded Systems & AI)

Personal data:

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Year of birth 1986
Nationality German



Overview

Expert in the verification of complex embedded hardware/software systems using Hardware-in-the-Loop (HiL). Extensive experience in CI/CD test automation, module integration, and agile software quality assurance. Experienced in implementing various phases according to ASPICE (HWE.2, HWE.3, SWE.5, SYS.4). My focus is on fostering innovation in high-performance engineering environments.

Work Experience

2009 – till now **Total 11.2 years' experience, Worked in various roles Hardware and Software Development / Test Engineer, In-Vehicle tests, Regensburg Germany**

Job Title: Software Engineer/Test Engineer – HIL (VW)
Company: Manu Dextra/Vitesco Technologies,
Location: Regensburg, Deutschland
07.2021 – 03.2024

- Maintain and upgrade HIL rigs (Vector VT System, RT Rack) for multiple ECUs.
- Perform advanced troubleshooting of hardware and software issues to minimize test bench downtime.
- Conducted manual and automated regression tests using LabView TestStand.
- Execute and Validation of software downloads (SWDL) on EMS ECUs.
- Collaborate with development and test teams to ensure test environment readiness.
- Maintaining and Updating python scripts for signal simulation and fault injection.
- Manage hardware inventory, calibration schedules, and system documentation.
- **Process Improvement:** Contribute to the continuous improvement of HIL testing methodologies, processes, and tools to enhance efficiency and test coverage.

Job Title: Software Engineer – Vehicle Testing
Company: Manu Dextra/Continental Automotive,
Location: Regensburg, Deutschland
08.2019 – 06.2021

- **In-Vehicle and Lab Testing:** Conducting manual and automated vehicle tests on the ID.3 platform, as well as lab tests to validate gateway requirements and ensure software stability for each sprint version.
- **Software Integration (ICAS1):** Responsible for software module integration within an Agile team (Jira) for the VW Gateway Control Unit (ICAS1 / ID.3).
- **CI/CD Integration:** Established and managed automated integration testing setups using Jenkins for continuous integration (CI/CD) workflows.
- **Software Flashing & Diagnostics:** Performed software flashing on hardware for every new release using ICAS1 debuggers, VW ODIS diagnostic tools, and custom Python scripts.
- **Release Management:** Coordinated with cross-site teams to integrate software packages from multiple sources for final delivery to the System Integration team.

- **Testing & Verification:** Conducted manual and automated regression tests on test benches and in-vehicle to validate gateway requirements for every sprint release.

Job Title:
Company:
Location:
02.2018 – 04.2019

IPN Brainpower/Continental Engineering Services, Nürnberg, Deutschland
Role: Software Engineer/ Test Engineer - ECU Verification

- **Test Execution:** Execute both automated and manual test procedures on the HIL system, covering nominal operation, boundary conditions, and complex scenarios (powertrain control logic).
- **Fault Injection Testing:** Design and execute tests using the **Fault Insertion Unit (FIU)** to simulate electrical failures (e.g., short to battery/ground, open circuits) and verify the ECU's diagnostic and fault-handling capabilities.
- **Data Analysis and Reporting:** Collect, analyse, and interpret large volumes of test data to identify software or hardware anomalies. Perform **root cause analysis (RCA)** and generate detailed test reports and defect tickets (e.g., in Jira or other issue tracking systems).

Job Title:
Company:
Location:
07.2017 – 12.2017

Conti Temic microelectronics GmbH, Body & Security, Markdorf , Germany
Role: Master Thesis Candidate

- **Over-the-Air (OTA) updates** for automotive refer to the process of wirelessly delivering and installing new software or **firmware** from a manufacturer's connected server (the source endpoint) directly to the service center onboard computers (the destination endpoint).

Job Title:
Company:
Location:
01.2017 – 06.2017

ADC Automotive Distance Control Systems GmbH, Lindau/ Ulm, Germany
Role: Student Intern – Testing

- **Test and Verification Automation:** Developed Python scripts to automate repetitive tasks in vehicle component testing and validation pipelines, ensuring adherence to strict automotive standards.

Job Title:
Company:
Location:
05.2011 – 03.2014

Robert Bosch Engineering and Business Solutions Limited, Coimbatore, India
Role: Senior Hardware Engineer/Senior HIL Test Engineer – Power Train

- **Hardware Integration:** Leading the integration and validation of electronic control units (ECUs) for diesel and gasoline powertrains.
- **OEM Collaboration:** Close coordination with Bosch GmbH and international OEMs (Fiat/Chrysler) to define technical hardware specifications.
- **Verification & Testing:** Conducting functional tests as well as EMC and environmental tests in accordance with stringent automotive quality standards.
- **Fault Analysis:** Root cause analysis and hardware debugging to resolve complex system integration problems.

Job Title:
Company:
Location:
07.2009 – 04.2011

McML Protection Technologies Pvt. Ltd.,Hyderabad, Indien
Role: Engineer R&D

- Design and implement initial proof-of-concept (PoC) hardware prototypes and development boards.
- Develop and validate critical firmware and drivers (e.g., C/C++) for new embedded hardware platforms.
- Set up and maintain experimental test benches and measurement equipment required for high-precision R&D validation.
- Perform rigorous testing (unit, integration, system-level) and debugging to resolve complex, low-level hardware/software synchronization issues.
- Conduct performance benchmarking and optimization of embedded algorithms to meet target KPIs (e.g., latency, throughput, energy consumption).

- Hobbies**
- Passionate about self-hosting open-source solutions, actively learning, experimenting, and deepening understanding of networking, server management, data privacy, and DevOps concepts
 - Previously developed Android apps, exploring mobile ecosystems and UI design

Education

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| 10.2024 – till now | PG Program in – Data Science and Artificial Intelligence, ExcelR , India |
| 04.2014 - 02.2018 | Master of Science – Automotive Software Engineering, Technische Universität Chemnitz, Germany |
| 04.2004 - 05.2008 | Bachelor of Technology in Electrical Engineering, SVEC, JNTU, India |

Technical skills and Competences

- **Core Competencies:** Automotive Embedded Systems, Software Development Process, Software Construction, Integration, Debugging, Analytical Thinking, Problem Solving, System and Product Know-how, Ability to Abstract
- **Architecture & Standards:** ISTQB standards Foundation Level Course Completion Certificate by TM Square, ASPICE, Functional Safety, Embedded Operating Systems.
- **Processes & Methodologies:** Software Verification & Validation, Change Management, Software Configuration Management, Continuous Integration, Agile Methodologies and Tools, IBM DOORS, Automotive SPICE.
- **Networking & Protocols:** BUS/Network Systems (CAN, UDS), Microcontrollers & ASICs Architecture, Serializer.
- **Languages & Tools:** C, C++, Python, Java, Kotlin, Cmake, Matlab Simulink, Autosar Builder, Elektrobit Tresos, Git/Gitlab, Jira. **DevOps Technologies:** CI/CD: AWS, Azure, Terraform(IaS), Ansible, Docker, Git, Gerrit, GitLab, Jira, Jenkins and Kubernetes (Ongoing Bootcamp)
- **Currently Exploring:** Gen AI, AI in Testing

Soft Skills

- **Leadership & Mentorship:** Successfully built and led Linux teams and technical communities.
- **Collaboration & Communication:** Engaged with global stakeholders and cross-functional teams.
- **Adaptability & Problem-Solving:** Swiftly tackled complex integration and debugging challenges.

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| Languages | English - Fluent,
German - B1 abgeschlossen, B2 aktuell in Arbeit
Telugu - Mother tongue |
| Driving License | Class -BE (since 2017) |